Printed Pages: 02 **Sub Code: KMBNIT-02**

Paper Id: 232347 **Roll No.** | | | | | | | | | | |

MBA
(SEM III) THEORY EXAMINATION 2022-23
AI AND ML FOR BUSINESS

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a. Define machine learning.
b. Give light on the term "artificial intelligence".
c. Brief about artificial neural network.
d. What is genetic algorithm?
e. Write K- Nearest Neighbour Algorithm.
f. What is supervised learning?
g. Define deep learning?
h. What do you understand by the term "perceptron"?
i. What is reinforcement learning?
j. What is Bayesian Network?

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a. Describe and discuss the issues of knowledge representation.
b. Enumerate the metrics for evaluating linear model.
c. Explain the density based method OPTICS of unsupervised learning.
d. Demonstrate the implementation of perceptron algorithm for AND Logic gate with bipolar inputs and targets.
e. Explain markov model of decision process.

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a. Elaborate on different types data sources for artificial intelligence.
b. Discuss techniques of knowledge representation in AI.

4. Attempt any *one* part of the following: **10 x 1 = 10**

a. Discuss working of Support Vector Machine Algorithm.
b. Describe how supervised learning is applied in solving business problem related to marketing.

5. Attempt any *one* part of the following: **10 x 1 = 10**

a. Explain hierarchal clustering in machine learning and discuss working of agglomerative and dendrogram.

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b. Discuss the concept of K-mean clustering with suitable example.

6. Attempt any *one* part of the following: 10 x 1 = 10

a. For the following given transaction data set, generate rules using Apriori Algorithm. Consider the values as Support = 50% and Confidence = 75%.

Transaction ID	Items Purchased
1	Bread, Cheese, Egg, Juice
2	Bread, Cheese, Juice
3	Bread, Milk, Yogurt
4	Bread, Juice, Milk
5	Cheese, Juice, Milk

b. Mention and discuss types of layers in convolution.

7. Attempt any *one* part of the following: 10 x 1 = 10

a. Explain the Q function and Q Learning Algorithm.
b. Explain Q learning algorithm assuming deterministic rewards and actions?

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